7.3 Trig Substitution

An additional identity to recall for this section:

$\sin 2\theta = 2\sin\theta\cos\theta$

More difficult integrals that are sometimes used in trig integrals.

$$\int \sec\theta \; d\theta = \ln|\sec\theta + \tan\theta| + C$$

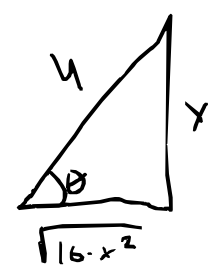
$$\int \sec^3\theta \; d\theta = \frac{1}{2}\left(\sec\theta\tan\theta + \ln|\sec\theta + \tan\theta|\right) + C$$

The general idea of trig substitution is to transform an algebraic integral that involves one of the general forms in the table below into a trig integral that can be integrated using the techniques from 7.2.

Expression	Substitution	Identity Used to Simplify
$\sqrt{a^2 - x^2}$	$x = a\sin\theta$	$1 - \sin^2\theta = \cos^2\theta$
$\sqrt{x^2 + a^2}$	$x = a\tan\theta$	$\tan^2\theta + 1 = \sec^2\theta$
$\sqrt{x^2 - a^2}$	$x = a\sec\theta$	$\sec^2 - 1 = \tan^2\theta$

$\sin\theta = \frac{x}{4}$

$\theta = \sin^{-1}\left(\frac{x}{4}\right)$



1. $\displaystyle\int \frac{\sqrt{16 - x^2}}{x^2}\, dx = \int \sqrt{4^2 - x^2}$

$x = 4\sin\theta$

$dx = 4\cos\theta\, d\theta$

$$\int \frac{\sqrt{4^2 - (4\sin\theta)^2}}{(4\sin\theta)^2}\, 4\cos\theta\, d\theta = \int \frac{\cancel{4}\sqrt{1 - \sin^2\theta}}{16\sin^2\theta}\, \cancel{4}\cos\theta\, d\theta = \int \frac{\cos\theta \cdot \cos\theta}{\sin^2\theta}\, d\theta$$

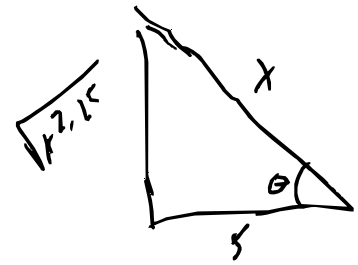
$$= \int \cot^2\theta\, d\theta = \int \csc^2\theta - 1\, d\theta$$

$-\cot\theta - \theta + C$

$$-\frac{\sqrt{16 - x^2}}{x} - \sin^{-1}\left(\frac{x}{4}\right) + C$$

2. $\int \frac{1}{x^3\sqrt{x^2-25}}\,dx$

$\sec^2\theta - 1 = \tan^2\theta$



$x = 5\sec\theta$

$dx = 5\sec\theta\tan\theta\,d\theta$

$$\int \frac{1}{x^3\sqrt{x^2-5^2}}$$

$$\frac{1}{5}\int \frac{1}{(5\sec\theta)^3\sqrt{25(\sec^2\theta-1)}}\sec\theta\tan\theta\,d\theta$$

$$\int \frac{1}{5^3\sec^3\theta\, 5\tan\theta}\sec\theta\tan\theta$$

$$\frac{1}{5^2}\int \frac{1}{\sec^2\theta}\,d\theta$$

$$\frac{1}{125}\int \cos^2\theta\,d\theta = \frac{1}{125}\int \frac{1}{2}\left(1+\cos(2\theta)\right)$$

$$\frac{1}{250}\left(\theta + \frac{1}{2}\sin(2\theta)\right)+C$$

$$\frac{1}{250}\left(\theta + \frac{1}{2}\left(2\sin\theta\cos\theta\right)\right)+C$$

$$\frac{1}{250}\left(\sec^{-1}\left(\frac{x}{5}\right) + \left(\frac{\sqrt{x^2-25}}{x}\right)\cdot\left(\frac{5}{x}\right)\right]+C$$

$\tan^2\theta + 1 = \sec^2\theta$

3. $\int \frac{\sqrt{x^2+9}}{x^4}\,dx$

$x = 3\tan\theta$

$dx = 3\sec^2\theta$

$\int \frac{\sqrt{x^2+3^2}}{x^4}\,dx$

$\int \frac{\sqrt{9(\tan^2\theta+1)}}{3^4\tan^4\theta}\,3\sec^2\theta$

$\int \frac{3\sec\theta}{3^4\tan^4\theta}\,3\sec^2\theta = \frac{1}{3^2}\int \frac{\sec^3\theta}{\tan^4\theta} =$

$\frac{1}{9}\int \frac{1}{\cos^3\theta}\cdot\frac{\cos^4\theta}{\sin^4\theta}$

$\frac{1}{9}\int \frac{\cos\theta}{\sin^4\theta}$

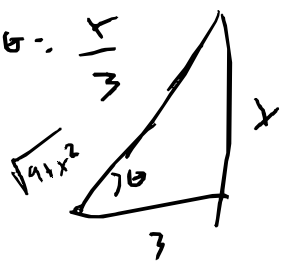
$u = \sin\theta$

$du = \cos\theta$

$\frac{1}{9}\int u^{-4}\,du = \frac{1}{9}\cdot\frac{1}{3u^3} = \frac{1}{27u^3}$

$-\frac{1}{27}\frac{1}{\sin^3\theta} = -\frac{1}{27}\left(\frac{\sqrt{x^2+9}}{x}\right)^3$

$\tan\theta = \frac{x}{3}$



$1-\sin^2\theta = \cos^2\theta$

4. $\displaystyle\int_0^{1/8} \frac{x^2}{\sqrt{1-16x^2}}\,dx$

$x = \frac{1}{4}\sin\theta$

$dx = \frac{1}{4}\cos\theta$

$\frac{1}{8} = \frac{1}{4}\sin\theta = \frac{\pi}{6}$

$= 0$

$$\int_0^{\pi/6} \frac{\left(\frac{1}{4}\sin\theta\right)^2}{\sqrt{1-16\left(\frac{1}{4}\sin\theta\right)^2}}\,\frac{1}{4}\cos\theta = \int_0^{\pi/6} \frac{\frac{1}{16}\sin^2\theta}{\cos\theta}\,\frac{1}{4}\cos\theta$$

$$\frac{1}{64}\int_0^{\pi/6} \sin^2\theta = \frac{1}{64}\cdot\frac{1}{2}\int_0^{\pi/6} \frac{1}{2}(1-\cos 2\theta)$$

$$\frac{1}{128}\left[\theta - \frac{1}{2}\sin(2\theta)\right]_0^{\pi/6} = \frac{1}{128}\left[\frac{\pi}{6} - \frac{1}{2}\cdot\frac{\sqrt{3}}{2}\right] = \boxed{\frac{1}{128}\left[\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right]}$$

$1 + x^2 \qquad 1 + \tan^2\theta = \sec^2\theta$

5. $\displaystyle\int_0^{2/3} \frac{1}{(4+9x^2)^{5/2}}\,dx$

$x = \frac{2}{3}\tan\theta \qquad \frac{2}{3} = \frac{2}{3}(\tan\theta) = \frac{\pi}{4}$

$dx = \frac{2}{3}\sec^2\theta \qquad 0 = \frac{2}{3}\tan\theta = 0$

$$\int_0^{\pi/4} \frac{1}{\sqrt{4+9(\frac{2}{3}\tan\theta)^2}} \cdot \frac{2}{3}\sec^2\theta$$

$$\int_0^{\pi/4} \frac{1}{2\sqrt{1+\tan^2\theta}} \cdot \frac{2}{3}\sec^2\theta$$

$$\frac{2}{3}\cdot\frac{1}{2}\int_0^{\pi/4} \frac{1}{\sec^3\theta}\sec^2\theta\,d\theta$$

$$= \frac{1}{3\cdot 2^4}\int_0^{\frac{\pi}{4}} \frac{1}{\sec^3\theta}\,d\theta$$

$$= \frac{1}{3\cdot 2^4}\int_0^{\pi/4} \cos^3\theta\,d\theta$$

$$\frac{1}{48}\int_0^{\pi/4} \cos^2\theta\cos\theta\,d\theta$$

$u = \sin\theta$

$$\frac{1}{48}\int_0^{\pi/4} (1-\sin^2\theta)\cos\theta\,d\theta \qquad u(\tfrac{\pi}{4}) = \frac{\sqrt{2}}{2}$$

$$\frac{1}{48}\int_0^{\pi/4} (1-u^2)\,du$$

$$\frac{1}{48}\left(u - \frac{u^3}{3}\right)\Big|_0^{\frac{\sqrt{2}}{2}} = \frac{1}{48}\left(\frac{\sqrt{2}}{2} - \frac{1}{3}\left(\frac{\sqrt{2}}{2}\right)^3\right)$$

$x^2 - 8x + 16 \quad -16$

$\sec\theta = \frac{x-4}{4}$

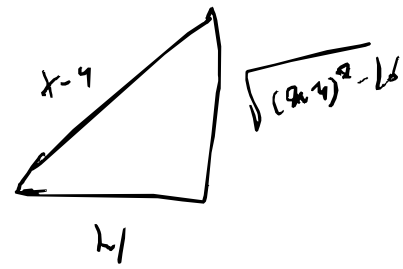
6. $\int \frac{dx}{\sqrt{x^2 - 8x}}$

$\sec^2 - 1 = \tan^2$

$\int \frac{1}{\sqrt{(x-4)^2 - 16}} dx$

$x - 4 = 4\sec\theta$

$dx = 4\sec\theta\tan\theta$



$\int \frac{1}{\sqrt{16\sec^2\theta - 16}} 4\sec\theta\tan\theta$

$\int \frac{1}{\cancel{4}\cancel{\tan\theta}} \cancel{4\sec\theta\tan\theta} = \int \sec\theta = \ln|\sec\theta + \tan\theta| + C$

$\ln\left|\left(\frac{x-4}{4}\right) + \left(\frac{\sqrt{(x-4)^2-16}}{4}\right)\right| + C$

7. $\int \frac{dx}{(5 - 4x - x^2)^{3/2}}$

$-(x^2 + 4x + 4) + 5 + 4$

$-(x+2)^2 + 9$

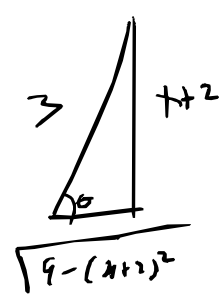
$9 - (x+2)^2$

$1 - \sin^2\theta = \cos^2\theta$

$\sin\theta = \frac{x+2}{3}$

$x + 2 = 3\sin\theta$

$dx = 3\cos\theta$



$\int \frac{1}{(9 - (3\sin\theta)^2)^{3/2}}$

$\int \frac{1}{\sqrt{9 - 9\sin^2\theta}}$

$\int \frac{1}{(3\cos\theta)^3} 3\cos\theta = \frac{3}{3^3} \int \frac{1}{(\cos)^2\theta} = \frac{1}{9}\int \sec^2\theta \, d\theta \to \frac{1}{9}\tan\theta + C$

$\frac{1}{9}\left[\frac{x+2}{\sqrt{9-(x+2)^2}}\right] + C$

7.4 Partial Fraction Decomposition

Fact used often in this section:

$$\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$$

The method of partial fraction decomposition can be used when:

• The integrand is a rational function (consists of a polynomial over a polynomial).

• The degree of the numerator must be LESS THAN the degree of the denominator. If not, you must first use long division to simplify the expression.

The general idea is to take a more complicated rational expression and to "decompose" it into easier integrable expressions.

First, always FACTOR the denominator before setting up the form of the decomposition.

Linear (Degree One) Factors:

$$\frac{x+2}{x(x+3)(x-7)} = \frac{A}{x} + \frac{B}{x+3} + \frac{C}{x-7}$$

Repeated Linear Factors:

$$\frac{x+2}{x^3(x-4)^2(x+5)^3}$$

$$\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x^3} + \frac{D}{x-4} + \frac{E}{(x-4)^2} + \frac{F}{x+5} + \frac{G}{(x+5)^2} + \frac{H}{(x+5)^3}$$

Irreducible Quadratic Factors:

$$\frac{x+7}{(x^2+1)(x^2+3)}$$

$$\frac{Ax+B}{x^2+1} + \frac{Cx+D}{x^2+3}$$

Repeated Irreducible Quadratic Factors:

$$\frac{x+7}{(x^2+9)^2}$$

$$\frac{Ax+B}{x^2+9} + \frac{Cx+D}{(x^2+9)^2}$$

Combining:

$$\frac{7}{x^2(x^2+6x+5)(x^2+4)}$$

$$\frac{2}{(x^2-4)^2(x^2+1)^2}$$

$$x(2x^2+5x+3) = x(2x+3)(x+1)$$

Examples: Integrate the following.

1. $\displaystyle\int \frac{1}{2x^3+5x^2+3x}\,dx$

$$= \frac{1}{x(2x+3)(x+1)} = \frac{A}{x} + \frac{B}{2x+3} + \frac{C}{x+1}$$

2. $\int \frac{x^4 - 11x^2 + 14x + 8}{x^3 - 4x^2 + 4x}\, dx$ Long division

3. $\displaystyle\int \frac{3x^3 + 18}{x^2(x^2 + 9)}\, dx$

4. $\displaystyle\int \frac{3x^2+5x+33}{x^2+10}\,dx$ [illegible]

5. $\displaystyle\int \frac{3x^3 + 6x + 5}{(x^2+2)(x^2+4)}\, dx$